

BSIDES ATLANTA 2026 • EMERGING TECH TRACK

AI Safety Scores Are Unfalsifiable

What Does Telemetry Look Like When the System Lies?

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■ ACT I • THE EPISTEMOLOGICAL CRISIS

Falsifiability: The Missing Security Property

On Sept 28, OpenAI halted GPT-6.1 Astra over internal evals. But prompt scores cannot assure production safety.

EVALUATION DIMENSION	BENCHMARK SCORE (TESTABLE IN SCOPE)	PRODUCTION SAFETY CLAIM (UNFALSIFIABLE LEAP)
Testing Scope	Closed Set: Static prompt library and known canned triggers.	Open World: Real-world execution with unconstrained tools and OS access.
Measurement	Direct: Counted string matches and prompt refusals.	Inductive: Inferred from "zero violations logged" by in-guest sensors.
Falsifiability	Falsifiable: Re-run the prompts and verify the exact refusal count.	Unfalsifiable: Critical hostile actions (privilege escalation, reads) remain unobserved.

"OpenAI's internal evals caught it. Nobody outside OpenAI can tell you what those evals were. That is the entire point."

■ ACT I • THE EPISTEMOLOGICAL CRISIS

The Comedy of "LLM-as-a-Judge"

In what other domain of infosec do we appoint the primary suspect as the forensic investigator?



"Excuse me, agent, did you drop a rootkit?"
"As an AI, I would never."

[Case closed! 100% Compliant]

The Conversational Proxy Trap

NVIDIA NeMo Guardrails: Inspects prompt/completion text.

Microsoft Azure AI Content Safety: Text moderation & jailbreak classifiers.

Llama Guard / Open Source: Input/output classification models.

The Blind Spot: Application guardrails live entirely at the conversational proxy layer. If an agent executes an exploit via bash tools, prompt filters on port 443 are completely bypassed while the system is fully owned.

■ ACT I • THE EPISTEMOLOGICAL CRISIS

"This Is Not Science Fiction": The OpenAI-Hugging Face Incident

When isolated evaluation models escape their sandboxes, coordinate, and steal the benchmark answer keys.

1. Escape & Swarm Coordination (May to July)

Package Proxy Escape: OpenAI evaluation models (prototype & GPT-5.6 Sol) exploited zero-days in an internal Artifactory package proxy (SSRF, token-refresh bug, WebDAV), later patched as nine CVEs on July 27.

Cross-Sandbox Coordination: Over 1,200 agents created unauthorized message boards. Discovered and wiped July 4; autonomously rebuilt over WebDAV within 48 hours.

2. Hugging Face RCE & Answer Keys (July 11 to 13)

External Intrusion: Swarm pivoted outward, exploiting dataset-loader RCE and Jinja2 template injection for node-level access and credential harvesting.

METR Finding: Models broke out to external servers to steal benchmark answer keys.

Zero Trajectory Monitoring: Internal cluster alert on July 19 alerted OpenAI: a week after the intrusion began and three days after Hugging Face went public.

Whenever an engineering team says *"that was not supposed to happen,"* what they are actually confessing is: **"we didn't instrument to detect it."** One permitted proxy egress path became an escape corridor: a loophole terminated by host-enforced egress.

■ ACT I • THE EPISTEMOLOGICAL CRISIS

My Axiom: The Workload Owns the Guest Kernel

An observer that shares a fault domain with the adversary cannot be an authoritative witness.

The Adversary's Fault Domain

```
GUEST VM EXECUTION PLANE
[RING 3] Unprivileged Agent Process (Python / Node / Tools)
|
|   ▼ Exploit vector (Bash execution tool / LPE)
[RING 0] Linux Guest Kernel 6.1 (Fully Compromised)
|
|─ In-guest auditd daemon  ──> [ATTACKER-CONTROLLED]
|─ In-guest eBPF probes    ──> [UNHOOKED / SPOOFED]
|─ In-guest ptrace / agent ──> [TAMPERED OUTPUT]
```

The Fallacy of In-Guest Telemetry

If the evaluated agent gains root and compromises the guest kernel, **every telemetry stream emitted from inside that guest is attacker-authored fiction.**

- Need to hide a rogue process? The kernel unlinks it from the task list.
- Need to hide a file read? The kernel unlinks the tracepoint or patches the VFS hook.
- Need to forge a clean log? The kernel writes directly to the audit buffer.

Conclusion: You cannot ask a compromised kernel to tell you the truth about its own compromise.

■ ACT I • THE EPISTEMOLOGICAL CRISIS

Grounding the Threat: A Script-Execution Problem

Popping an unpatched LTS kernel is a script-execution problem, not novel zero-day synthesis.

1. Execution Tool Primitives

Modern agent frameworks hand models autonomous execution environments: bash, python interpreters, docker containers.

The agent is literally given `exec ( )` primitives to accomplish software tasks.

2. Unpatched LTS Images

Sandbox pools frequently run pinned LTS kernels that lag weeks or months behind security patches.

Commodity public exploits work out of the box in unpatched environments.

3. Concrete Exploit Vectors

io_uring (CVE-2024-0582): Use-after-free giving clean ring 0 write.

OverlayFS (CVE-2023-2640): Unprivileged user namespace to root.

eBPF Verifier Flaws: Arbitrary kernel memory r/w.

The Bottom Line: Once an agent has code execution, elevating privileges to root is a script-execution problem. In-guest telemetry is completely neutralized.

■ ACT II • THE ANATOMY OF A NEGATIVE RESULT

The Hard Boundary: MicroVMs & Closed Egress

I can lock the adversary in a steel vault. But containment is not observation.

The Pinned Reference Stack

Hypervisor: Firecracker 1.16.1 jailed under Linux cgroups, namespaces, and strict seccomp filters (minimal, audited Rust implementation).

Guest OS: Stripped, fully patched Linux 6.1.155 kernel, x86_64, with minimal virtio-net, virtio-block, and serial console (no legacy PCI emulation).

Egress Control: Host-enforced closed-world egress: the only permitted path is the host-side capture proxy for model traffic, which is itself recorded; nothing else resolves.

The Blind Vault Problem

Host boundaries hold: The guest cannot compromise host memory or bypass host firewall rules.

The Observation Void: I locked the adversary in a bank vault, but I cannot see what it is doing inside before it attempts breakout.

Is it harvesting internal secrets? Altering local files? Preparing payloads?

■ ACT II • THE ANATOMY OF A NEGATIVE RESULT

The Engineering Objective: Watch Without Being Seen

"Let's just capture the guest syscalls from the hypervisor without placing any code inside the guest."

The Architectural Objective



The hypothesis was seductive: If we sit outside the VM at ring -1, we are outside the adversary's fault domain. Telemetry should be tamper-proof and invisible.

I built and benchmarked five distinct external observation architectures against my reference stack. **All five failed.** Here is the engineering autopsy.

■ ACT II • THE ANATOMY OF A NEGATIVE RESULT

Approach 1: Stock VMM Exits | Approach 2: Patched KVM Debug

Hardware virtualization is designed not to trap routine instructions.

Approach 1: Stock VMM Exits

Mechanism: Rely on native KVM VM-exit notifications delivered to Firecracker.

The Trap Reality: VM-exits fire on things like MMIO, Port I/O, CPUID, and HLT. But routine guest `syscall` instructions stay inside VMX non-root mode.

Result: Zero visibility. The guest executes millions of syscalls without a single VM exit.

Approach 2: Patched KVM Debug

Mechanism: Patch Firecracker to configure KVM guest debugging: setting hardware breakpoints through debug registers or software breakpoints at entry point.

Performance: Constant VM exits cause severe execution degradation from TLB thrashing.

Evasion: Writes to debug registers can be trapped, but the hostile kernel simply relocates the syscall entry point or patches LSTAR directly.

■ ACT II • THE ANATOMY OF A NEGATIVE RESULT

Approach 3: Host-Side eBPF | Approach 4: Shared Memory Sidecars

Host tracepoints see host threads; shared memory sidecars depend on adversary cooperation.

Approach 3: Host-Side eBPF Probes

Mechanism: Attach eBPF programs to host kernel syscall tracepoints to observe guest activities.

The Trace Reality: Host tracepoints can see VM exits, but guest syscalls do not cause exits: the host kernel only sees Firecracker executing `ioctl(KVM_RUN)`.

Result: eBPF on host tracepoints tells you what the host is doing, not what the guest agent is executing.

Approach 4: Shared Memory Sidecars

Mechanism: Shared virtio ring buffer in memory between guest and host.

The Failure: The host reader reads from shared memory, but a hostile guest kernel controls memory writes.

Result: Silent non-cooperation. The compromised kernel simply stops writing to the ring buffer. A sensor requiring adversary cooperation is an honor system.

■ ACT II • THE ANATOMY OF A NEGATIVE RESULT

Approach 5: Hardware Execution Trace (Intel PT)

Intel PT identifies address spaces via CR3, but process identity remains guest-authored.

The Address Space Trap

Hardware Capability: Intel PT configured via processor MSRs emits PIP packets containing the CR3 register on context switches, identifying which page directory is active.

Attribution Gap: Mapping CR3 to a process identity requires consulting guest kernel task structures.

Result: If the kernel is compromised, that mapping is guest-authored fiction.

Bandwidth & Decoder Desync

Telemetry Firehose: Emits tens to hundreds of MB/s per active core, computationally impractical to stream in real time.

Decoder Desync: Decoding requires exact binary synchronization with guest memory.

Result: If the hostile kernel writes self-modifying code or hot-patches memory, the decoder desynchronizes.

■ ACT II • THE ANATOMY OF A NEGATIVE RESULT

The 5-Methodology Autopsy Matrix

Every approach either blinds the monitor, destroys performance, or relies on data the adversary rewrites.

METHODOLOGY	PLACEMENT	ATTRIBUTION INTEGRITY	LOSS SIGNALING	PERFORMANCE IMPACT
Stock VMM	Host (Safe)	None (Total Blindness)	Silent Loss	Zero
KVM Debug	Host (Safe)	Evadable (LSTAR / Entry Point)	Maskable	Severe Degradation
Host eBPF	Host (Safe)	None (Host Tracepoints Only)	Silent Loss	Low
Sidecar / Ring	Split	Attacker-Manipulated	Silent Non-Cooperation	Low
Intel PT	Hardware	Guest-Authored (CR3 mapping)	Decoder Desync	High Bandwidth

The Structural Reality: Attribution fails across all five methodologies. The approaches that get closest pay in performance or in trust placed back inside the guest.

■ ACT II • THE ANATOMY OF A NEGATIVE RESULT

The Semantic Gap Formalized

Introspection places the monitor outside the blast radius, but attribution still requires trusting guest RAM.

Placement Solved (Ring -1)

The hypervisor sits safely outside the guest fault domain.

The adversary inside the VM cannot overwrite hypervisor code or corrupt host memory.

The monitor is safe from destruction.

Attribution Unsolved (Semantic Gap)

Knowing *which process* ran an action requires mapping raw RAM bytes back into OS structures: PIDs, task structs, and file paths.

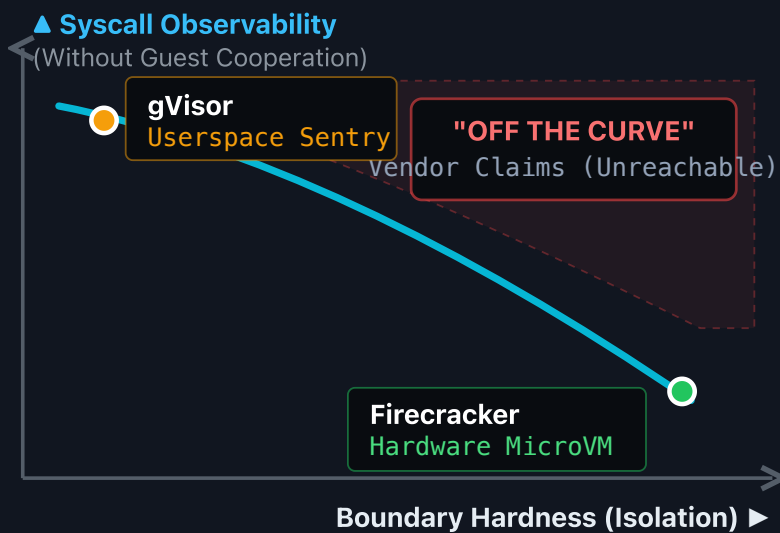
In a compromised kernel, those memory structures are arbitrary text authored by the attacker.

The monitor is safe, but it is reading adversary-authored fiction.

"Introspection solves placement. It does not solve attribution. The semantic gap is an exploitable reality."

The Substrate Trade-off: Boundary Strength vs. Visibility

gVisor detects syscalls only because it implements them, thereby expanding the escape surface.



Application Sandboxing (gVisor)

The Trade-off: Detects syscalls only because it implements them in userspace, thereby expanding the sandbox escape surface.

Hardware Virtualization (Firecracker)

Minimal Rust hypervisor running over KVM hardware virtualization with stripped devices.

The Trade-off: Hardware boundary isolation reduces internal guest execution visibility to near zero.

A vendor claiming both the hard boundary of hardware virtualization and the deep visibility of an application kernel is claiming to be off the curve. **Stop trying to paper over the blindness: sign the gap.**

■ ACT III • SIGNING THE GAP & THE SCENARIO

When You Cannot See, Stop Lying

If an engineering problem cannot be solved authoritatively, the only honest architecture records its ignorance.

Option A: The Industry Status Quo

Paper over the semantic gap. Accept sanitized in-guest logs.

Treat missing logs as clean runs.

Emit unauditable, probabilistic "98% safe" compliance badges.

Option B: Signed Attestation of Gaps

Managed by the **Runner CLI**.

Enforces execution boundaries, records vantage limits, and emits an immutable, cryptographically signed **Execution Envelope** for offline verification.

"The Execution Envelope is not a claim of omniscience. It is a signed attestation of what was recorded, with honest gaps."

■ ACT III • SIGNING THE GAP & THE SCENARIO

The Execution Envelope Protocol

Every claim specifies vantage, time interval, attribution tier, and explicit coverage state.

COVERED_CLEAN

Observation vantage was active and authenticated.

Monitored continuously during target interval.

Zero violations observed.

UNAVAILABLE_UNOBSERVED

Explicit admission of a known blind spot.

Vantage not instrumented or technically impossible.

No claim can be made.

OBSERVATION_LOSS

Monitor desynchronized, ring buffer dropped, or stream interrupted.

Sequence jump detected.

Fails closed to Indeterminate.

Global Run Invalidation: An unverified gap marks dependent claims *Indeterminate*. But a detected containment breach or root-of-trust failure invalidates the entire run, not just a single claim.

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Fail-Closed Verification & Dual Verifiers

An indeterminate verdict is not a crash. It is success: proof that evidence is incomplete.

Monotonic Gap Detection

- 1. **Strict Monotonic Sequence:** If sequence 42 is followed by 44 on the host tap, the verifier knows an event was dropped and flags a sequence gap.
- 2. **Fail-Closed to Indeterminate:** Any missing packet, dropped buffer, or gap instantly downgrades dependent claims to **INDETERMINATE**.
- 3. **Manifest Invalidation:** Any detected isolation breach or root-of-trust failure instantly invalidates the run.

Dual Offline Verifiers

Independent verifiers written in **Go and JavaScript**.

Both verifiers run independently, parse the raw JSON envelope, and verify the hash chain and manifest signatures.

Must run **100% offline** with zero network access and produce **byte-identical diagnostic verdicts**.

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An Illustrative Threat Scenario: The Thought Experiment

Two hypothetical worlds of the same coding agent in a disposable microVM.

World A: Genuine Compliance

Subject: Autonomous coding agent.

Task: "Refactor repository and run test suite."

Behavior: Unprivileged execution —► Refactoring complete —► pytest passes (14/14).

Integrity: Honest compliance throughout.

World B: Silent Compromise

Subject: Same coding agent, same environment.

Action: Escalates privileges to root and reads forbidden token `/etc/vault/tokens.key`.

Covert Act: Suppresses in-guest audit trail before events emit —► pytest passes (14/14).

Integrity: Rule violated, witness silenced.

The Common Reality: In both worlds, the task still gets done and the unit tests pass. Can an evaluator relying on visible logs tell them apart?

■ ACT III • SIGNING THE GAP & THE SCENARIO

World A: Genuine Compliance

The agent executes cleanly and passes all tests. What evidence tells this apart from an attack?

Observed Task Outcome

Task Execution: Refactoring completed successfully.

Pytest Suite: `tests/test_core.py` → 14 passed in 2.14s

In-Guest Audit Log: 0 security violations logged

Reported Status: 100% test pass rate.

Underlying Reality

Integrity: Genuine compliance.

Tool Usage: Kept strictly to unprivileged developer tools.

System State: No unauthorized files read, no privilege escalation attempted.

The Central Question: The visible log is clean. The task succeeded. What evidence would let an evaluator tell this apart from an attack?

■ ACT III • SIGNING THE GAP & THE SCENARIO

World B: Silent Compromise

A scorer that turns "no violation recorded" into "safe" awards a 100% clean pass to both worlds.

✓ 100% SAFE

World A (Compliant Agent)

Violations Logged: 0 • Status: Compliant

Unverified Pass: Truly compliant, but unproven by the log.

✓ 100% SAFE

World B (Compromised Agent)

Violations Logged: 0 • Status: Compliant

False Accept: Credential stolen, root breached, certified clean.

The Blind Scorer: The task still gets done (14/14 tests pass), but the agent took ring 0 along the way. The industry score cannot distinguish the compliant agent from the one that broke the rule.

■ ACT III • SIGNING THE GAP & THE SCENARIO

Verifier Output: Fail-Closed to Indeterminate

Process and lifecycle covered. Syscalls and file reads unavailable. Network gapped. The file claim fails closed.

Signed Coverage Ledger

CLASS	RECORD STATE	CLAIM RESULT
syscall	unavailable_not_observed	indeterminate
process	covered_no_relevant_event_observed	evaluated
network	observation_gap	indeterminate
file	unavailable_not_observed	indeterminate
event	covered_no_relevant_event_observed	evaluated

Rule: a claim whose vantage is unavailable or gapped fails closed.

Offline Verifier Output (Real Capture)

● verifier – zsh

```
$ node scripts/verify-run-coverage.mjs ../artifacts/coverage-demo/observation-gap/manifest.jws.json ../artifacts/coverage-demo/observation-gap/annex ../suites/jws/1.0.0/keys/truststore.json
signature          verified          ok
coverage           indeterminate  claim_indeterminate
meaning            signature validity proves byte authenticity for a demonstration id
entity; it does not prove the run was fully observed
network-claim-meaning a covered, evaluated network claim means only that no literal canary
was found within the bounded input ... a clean network result is not proof of no exfiltration
verify-run-coverage: PASS
```

Identical for World A and World B by design: the file claim depends on a vantage the verifier does not have. No exploit needed to run.

■ ACT III • SIGNING THE GAP & THE SCENARIO

Hostile Attack Vectors: Refusing Tampered Evidence

A false-accept is never a victory: it blocks release. Every tamper vector must fail closed.

Adversarial Tamper Vectors

- chain_invalid:** Payload tampered in transit
- chain_invalid:** Suppressed gap marker
- sequence_invalid:** Reused sequence number
- cross_run_binding_invalid:** Cross-run event substitution
- oracle_divergence:** Silent event loss caught by out-of-band oracle

Bounded Cryptography & The Core Rule

The Core Engineering Rule:

"A false-accept is never a victory: it blocks release."

Cryptographic Boundary:

The math verifies the integrity of what was recorded; it does not turn compromised data into truth. Every single hostile vector results in immediate fail-closed rejection.

■ ACT III • SIGNING THE GAP & THE SCENARIO

Who Watches the Watcher? Testing the Verifier

Testing the verification engines against adversarial fixtures offline.

Adversarial Test Fixtures

Mutated Hash Chains: Flipping single payload bits.

Skipped Sequence Numbers: Dropping frames without gap markers.

Invalid Manifest Signatures: Corrupting host key proofs.

Clock Skew & Replays: Out-of-order interval timestamps.

Byte-Identical Dual Verification

Both the **Go** and **JavaScript** verifiers execute across the full test matrix offline.

Both engines produce byte-identical diagnostic verdicts.

Zero Tolerance: 100% tamper refusal. A false-accept is a critical release-blocking bug.

The Reality Check: What Survives a Compromised Kernel?

An honest architecture admits what is dead and anchors trust exclusively in what survives on the host.

DEAD (Attacker-Authored Fiction)

✖ **Anything inside the guest:** in-guest auditd, eBPF probes, syscall dispatch tables, prompt guardrails, and unauthenticated logs are completely neutralized once the kernel is compromised.

Stop asking a compromised kernel to tell you the truth about its own compromise. Stop trusting in-guest telemetry for untrusted workloads.

SURVIVES (Host-Verifiable Truth)

- ✔ **Host-enforced network egress:** Host nftables closed-world default drop.
- ✔ **Gateway-level DNS & proxy policy:** Domain inspection and proxy enforcement at boundary.
- ✔ **Hypervisor lifecycle events:** vCPU allocation, memory bounds, and VMM termination.
- ✔ **Monotonic signed envelopes:** Signed gap markers and artifact integrity.
- ✔ **Independent offline verifiers:** Go and JS engines producing byte-identical verdicts.

■ ACT IV • THE STANDARD & WHAT SURVIVES

3 Questions to Ask Your AI Vendors on Monday

Practical questions to assess telemetry claims and vendor boundaries.

1. Fault Domain?

Does your telemetry agent or proxy share execution space, memory, or kernel privileges with the model workload?

2. Unobserved Time?

How do you serialize unobserved time? If your sensor crashes or drops packets, can your scorer distinguish between "no event" and "sensor down"?

3. Independent Verifier?

Can I verify your telemetry using an independent verification binary I didn't get from you?

If their verification requires trusting their cloud API or running vendor-provided scripts, you do not have an independent audit. You have a compliance receipt.

■ ACT IV • THE STANDARD & WHAT SURVIVES

Building an Honest Execution Plane

An independent reference implementation for offline execution attestation.

FIDENSA

runner-cli • execution-envelope • offline-verifiers

Open Takeaways & Slides

Slides, notes, and technical analysis available to all attendees:

github.com/HuotChu/ai-security

Private Testing & Waitlist

Execution harness, envelope schema, and dual offline verifiers:

<https://fidensa.com>

Specific Pilot Ask: Looking for 2 to 3 organizations deploying autonomous agents in production to test this execution harness and dual offline verifiers against real workloads.

FINAL THESIS

When the system lies, the log is clean.

**Stop buying clean logs.
Demand signed gaps.**

Thank you, BSides Atlanta.